

Task 3: Secure Coding Review (C++)

Description

This project is a secure coding review of a C++ login application. The purpose of this task is to identify common security vulnerabilities in code and improve the application using secure programming practices.

The project includes a vulnerable version of the application, a secured version, and a security analysis report.

Project Files

- `vulnerable_login.cpp`
Contains the original application with security weaknesses.
- `secure_login.cpp`
Improved version with security fixes.
- `Security_Report.md`
Contains vulnerability analysis and recommendations.

Vulnerabilities Identified

1. Hardcoded Passwords

Problem:
Credentials were directly written inside the source code.

Risk:
Attackers can view the code and steal login information.

Solution:
Avoid storing secrets in code and use secure storage methods.

2. Plain Text Password Storage

Problem:
Passwords were stored without protection.

Risk:

A data leak can expose user passwords.

Solution:

Use password hashing techniques.

3. Missing Input Validation

Problem:

User input was not properly checked.

Risk:

Invalid or malicious input may cause security problems.

Solution:

Validate user input before processing.

Security Improvements

- Added password hashing
- Improved authentication logic
- Added input validation
- Removed insecure coding practices

Technologies Used

- C++
- Secure Coding Practices
- Cybersecurity Concepts

Learning Outcome

Through this project, I learned:

- How to review source code for vulnerabilities
- Common software security issues
- Secure authentication techniques
- How to improve application security

Project Type

Cyber Security Internship Task - CodeAlpha

Vulnerable C++ Application

C++ Online Compiler

```
main.cpp
1 #include <iostream>
2 #include <string>
3
4 using namespace std;
5
6
7 int main()
8 {
9     string username;
10    string password;
11
12
13    // Hardcoded credentials
14    string correctUser = "admin";
15    string correctPass = "admin123";
16
17
18    cout << "Username: ";
19    cin >> username;
20
21
22    cout << "Password: ";
```

Output

```
Username: admin
Password: admin123
Login Successful

=== Code Execution Successful ===
```

Secure C++ Version

C++ Online Compiler

```
main.cpp
1 #include <iostream>
2 #include <string>
3 #include <functional>
4
5 using namespace std;
6
7
8 // Simple password hashing function
9
10 string hashPassword(string password)
11 {
12     hash<string> hasher;
13
14     return to_string(hasher(password));
15 }
16
17
18
19 int main()
20 {
21
22     string username;
```

Output

```
Username: admin
Password: admin123
Login Successful

=== Code Execution Successful ===
```

Activate Windows
Go to Settings to activate Windows.